



OPEN Efficacy of combined undenatured type II collagen and hydrolysed collagen supplementation in knee osteoarthritis: a randomised controlled trial

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Collagen supplements have gained significant market popularity, with numerous products claiming to improve knee osteoarthritis symptoms. Previous clinical studies have primarily evaluated individual collagen formulations, with most being industry-sponsored, which raises potential bias concerns. This study aimed to evaluate the efficacy of combined undenatured type II collagen (UC-II) and hydrolysed collagen supplementation in patients with knee osteoarthritis. In this randomised, double-blind, placebo-controlled trial, 68 patients with knee osteoarthritis were divided into two groups: one receiving collagen supplements (tablets containing hydrolysed collagen and UC-II) and the other receiving a placebo. Clinical outcomes, including pain intensity (Verbal Numerical Rating Scale), functional outcomes (Knee Injury and Osteoarthritis Outcome Score [KOOS]), rescue medication consumption, and patient satisfaction, were assessed at baseline, 2, 6, and 12 weeks. Baseline demographic characteristics were comparable between the collagen and placebo groups. Both groups showed significant improvements in pain intensity ($p < 0.001$) and KOOS scores over time. However, no significant differences were observed between the groups in terms of pain intensity, functional outcomes, rescue medication consumption, or patient satisfaction ($p > 0.05$). No adverse effects were reported in either group. Combined UC-II and hydrolysed collagen supplementation over 12 weeks did not demonstrate superior efficacy compared with that of the placebo in managing knee osteoarthritis symptoms. Further research is required to explore the potential long-term benefits of combined collagen supplementation.

Clinical Trial Registration: ClinicalTrials.gov (NCT06229951): registered on January 16, 2024.

Keywords Undenatured type II collagen, Hydrolysed collagen, Knee osteoarthritis, Randomised trial

Knee osteoarthritis (OA), which is characterised by the progressive degradation of articular cartilage, exhibits a significantly increasing prevalence owing to population aging and rising obesity rates¹. This condition substantially affects the quality of life of patients and poses significant socioeconomic implications^{2,3}. Current treatment modalities encompass a spectrum of interventions, ranging from behavioural modifications and pharmacological interventions to intra-articular injections and surgical procedures^{4,5}.

Collagen supplements have become increasingly popular in recent years, with a wide range of products claiming to improve the symptoms of knee OA. Collagen, the primary structural protein in articular cartilage, provides joint structure and elasticity⁶. Various forms of collagen supplements have emerged in clinical practice, including porcine collagen peptides, bovine collagen peptides, undenatured type II collagen (UC-II), and hydrolysed collagen from chicken cartilage⁷. Mechanistic studies have demonstrated that collagen may stimulate cartilage regeneration by enhancing the synthesis of extracellular matrix macromolecules⁸. In addition, research has revealed its immunomodulatory effects, preventing joint damage through the induction and migration of regulatory T cells and production of anti-inflammatory cytokines^{9,10}.

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Clinical investigations of collagen supplementation in patients with OA have shown promising results, with several studies demonstrating improvements in pain reduction and knee function when using hydrolysed collagen or UC-II^{11–15}. These findings suggested the potential therapeutic benefits of collagen supplementation in managing knee OA symptoms.

However, previous clinical studies have evaluated individual collagen formulations, with distinct trials investigating either hydrolysed collagen or UC-II in patients with OA. Although these single-formulation studies reported improvements in pain and knee function, most were industry-sponsored, raising potential bias concerns. Furthermore, the heterogeneity in collagen types and dosages across studies complicates the interpretation of optimal supplementation protocols. To date, no study has investigated the potential synergistic effects of different collagen formulations. Given these limitations, we conducted this manufacturer-independent trial to evaluate the efficacy of combined hydrolysed collagen and UC-II compared with that of placebo in knee OA management.

Materials and methods

This randomised, double-blind, placebo-controlled trial was conducted at the university hospital between November 2023 and August 2024. The study protocol was approved by the institutional review board on September 29, 2023. The trial is registered with ClinicalTrials.gov (NCT06229951) and was registered on January 16, 2024. Written informed consent was obtained from all participants before enrolment. All methods were performed in accordance with the relevant ethical guidelines and regulations, including the Declaration of Helsinki and applicable local regulations.

Patients diagnosed with knee OA seeking treatment at the outpatient orthopaedic clinic were recruited. The diagnosis was confirmed via clinical examination and radiographic findings. Patients were eligible if they were aged 50–80 years, had predominant symptoms in the medial knee compartment, demonstrated Kellgren–Lawrence radiographic classification grades of 2 or 3, had knee varus angulation $< 10^\circ$, and were able to walk without assistive devices. Patients were excluded if they had a history of knee surgery; received intra-articular steroid injection within the past 6 months; had systemic arthritis, neurological or muscular impairment of the lower extremities, or kidney disease with glomerular filtration rate < 60 mL/min/1.73 m²; or had a history of allergy to meloxicam, celecoxib, or collagen.

After screening 87 patients, 68 eligible participants were enrolled and randomly allocated to either the collagen ($n = 34$) or placebo ($n = 34$) group (Fig. 1). Randomisation was performed using a computer-generated sequence with a block size of four. The allocation sequence was concealed using opaque, sealed, and numbered envelopes that were opened sequentially by an independent research assistant who was not involved in clinical care or outcome assessment after confirming participant eligibility. The participants, treating physicians, and outcome assessors were blinded to the group allocation throughout the study period.

Participants in the collagen group received daily supplement tablets containing 480 mg hydrolysed collagen and 20 mg UC-II (Bangkok Drug Co., Ltd., Bangkok, Thailand), whereas those in the placebo group received tablets containing maltodextrin. The dosing schedule for both consisted of two tablets administered before bedtime. All participants received standardised care, including patient education, muscle-strengthening exercises, and rescue medication (meloxicam [7.5 mg] or celecoxib [200 mg] for those allergic to meloxicam) as needed for pain management.

Follow-up visits were scheduled at baseline and at weeks 2, 6, and 12. A blinded independent research assistant performed all the assessments. The primary outcome measure was pain level, measured using the Verbal Numerical Rating Scale (VNRS). Secondary outcomes included knee function and quality of life assessed using the Knee Injury and Osteoarthritis Outcome Score (KOOS), rescue medication consumption, patient satisfaction measured using the VNRS, and monitoring of adverse effects.

All statistical analyses were performed using R software (version 4.4.1; R Foundation for Statistical Computing, Vienna, Austria). The Kolmogorov–Smirnov test was used to assess the normality of data distribution. Independent *t*-tests were used to compare normally distributed continuous variables (age, weight, height, and body mass index [BMI]) between groups. Categorical variables (sex, Kellgren–Lawrence classification, and history of diabetes) were analysed using the chi-square test. Non-parametric data, including pain scores (VNRS), KOOS subscales, rescue medication consumption, and patient satisfaction scores, were compared between groups using the Mann–Whitney U test. Statistical uncertainty was assessed using 95% confidence intervals for the differences between the two groups. A Generalized Estimating Equations (GEE) analysis was performed to examine the effect of time and group on pain scores (VNRS) over a 12-week period. An intention-to-treat analysis was performed, including all randomised participants regardless of protocol adherence or study completion. Statistical significance was set at $p < 0.05$.

Sample size calculation was performed based on previous data from Kumar et al.¹¹ To detect a clinically meaningful difference in pain scores between groups, assuming a two-tailed α of 0.05 and β of 0.10 (power = 0.90), 28 participants were required per group. The final sample size was increased to 34 participants per group ($n = 68$) to account for an anticipated dropout rate of 20%.

Results

The baseline demographic characteristics, including sex, age, weight, height, BMI, Kellgren–Lawrence classification, and history of diabetes, were comparable between the collagen and control groups (Table 1).

Pain intensity measured using the VNRS demonstrated significant improvement from baseline in both groups at all follow-up time points (2, 6, and 12 weeks; $p < 0.001$). However, between-group comparisons revealed no significant differences at any time point. (Table 2) A Generalized Estimating Equations analysis revealed significant reductions in VNRS pain scores over time in both groups. At the 12-week follow-up, the collagen

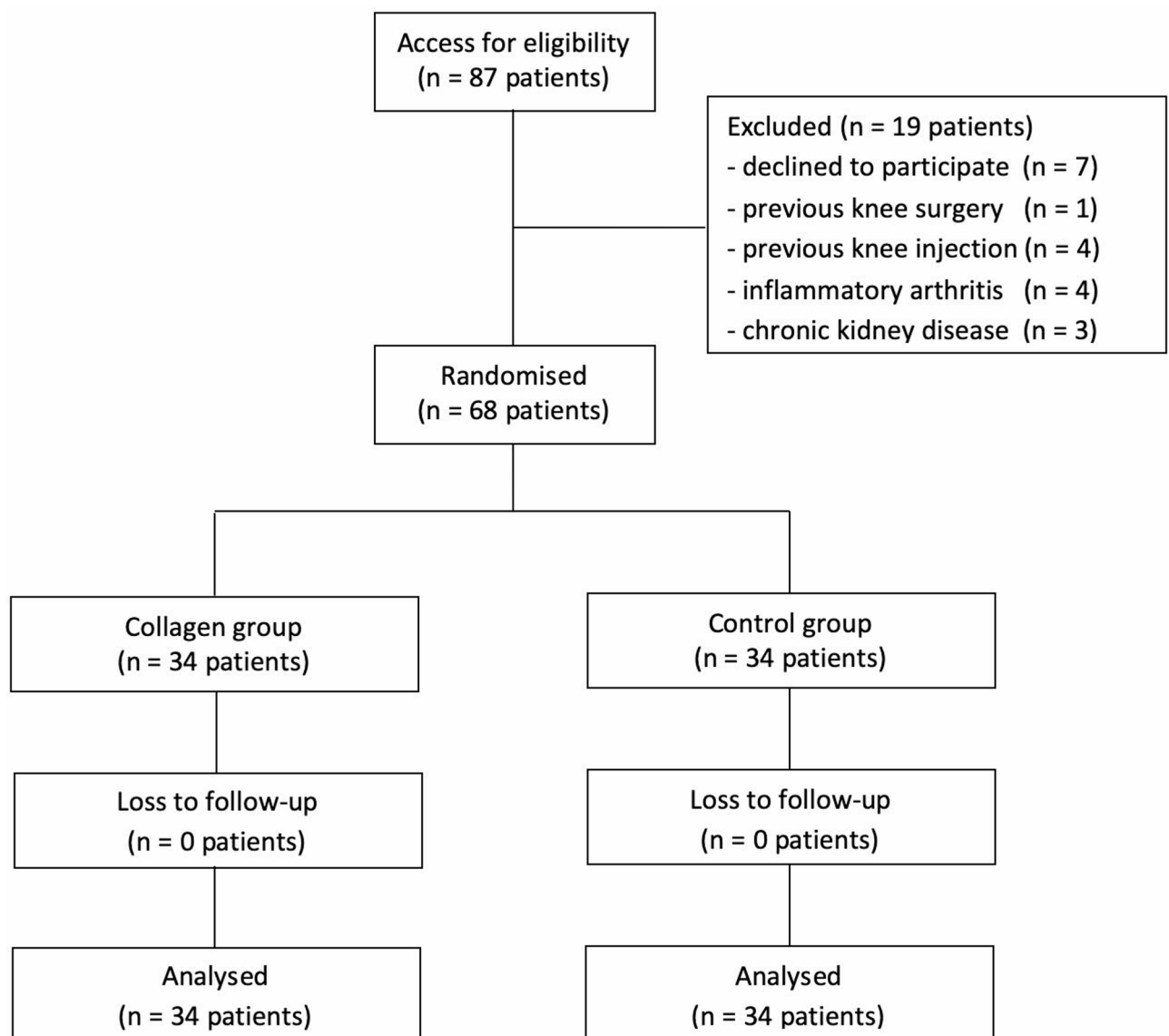


Fig. 1. Flow diagram of participant selection.

Characteristic	Collagen group n = 34	Control group n = 34	p value
Sex (male: female)	5:29	4:30	0.720
Age (years)	63.59 ± 7.11*	65.15 ± 6.61*	0.352
Weight (kg)	63.75 ± 11.04*	63.53 ± 9.04*	0.930
Height (cm)	156.69 ± 5.72*	154.74 ± 6.73*	0.201
BMI (kg/m ²)	26.01 ± 4.59*	26.51 ± 3.31*	0.602
KL classification (II: III)	23/11	23/11	1
Diabetes (%)	23.5%	23.5%	1

Table 1. Demographic data. *Values are expressed as the mean ± SD. BMI body mass index, KL Kellgren–Lawrence, SD standard deviation.

group showed a mean pain reduction of 2.21 points ($B = -2.21$, $p < 0.001$, 95%CI [-3.11, -1.30]), while the control group showed a reduction of 2.29 points ($B = -2.29$, $p < 0.001$, 95%CI [-3.33, -1.26]) compared to baseline. The difference in pain score reduction between the groups was minimal (0.08 points) and not statistically significant. (Fig. 2)

		Baseline	2 weeks	6 weeks	12 weeks
VNRS for pain	Collagen	5.5 (2)	4 (2.5)	3 (3)	3(3)
	Control	6 (2)	4 (2.3)	3 (3)	3 (3)
	p value	0.358	0.980	0.861	0.789
	95% CI	– 1, 0	– 1, 1	– 1, 1	– 1, 1
VNRS for satisfaction	Collagen	–	8 (3)	9 (2)	8.5 (1.3)
	Control	–	7 (3)	8 (3)	9 (2)
	p value	–	0.227	0.355	0.126
	95% CI	–	0, 2	0, 1	– 1, 0

Table 2. Comparison of verbal numerical rating scale (VNRS) for pain and satisfaction scores between collagen and control groups. *IQR* interquartile range, *CI* confidence interval.

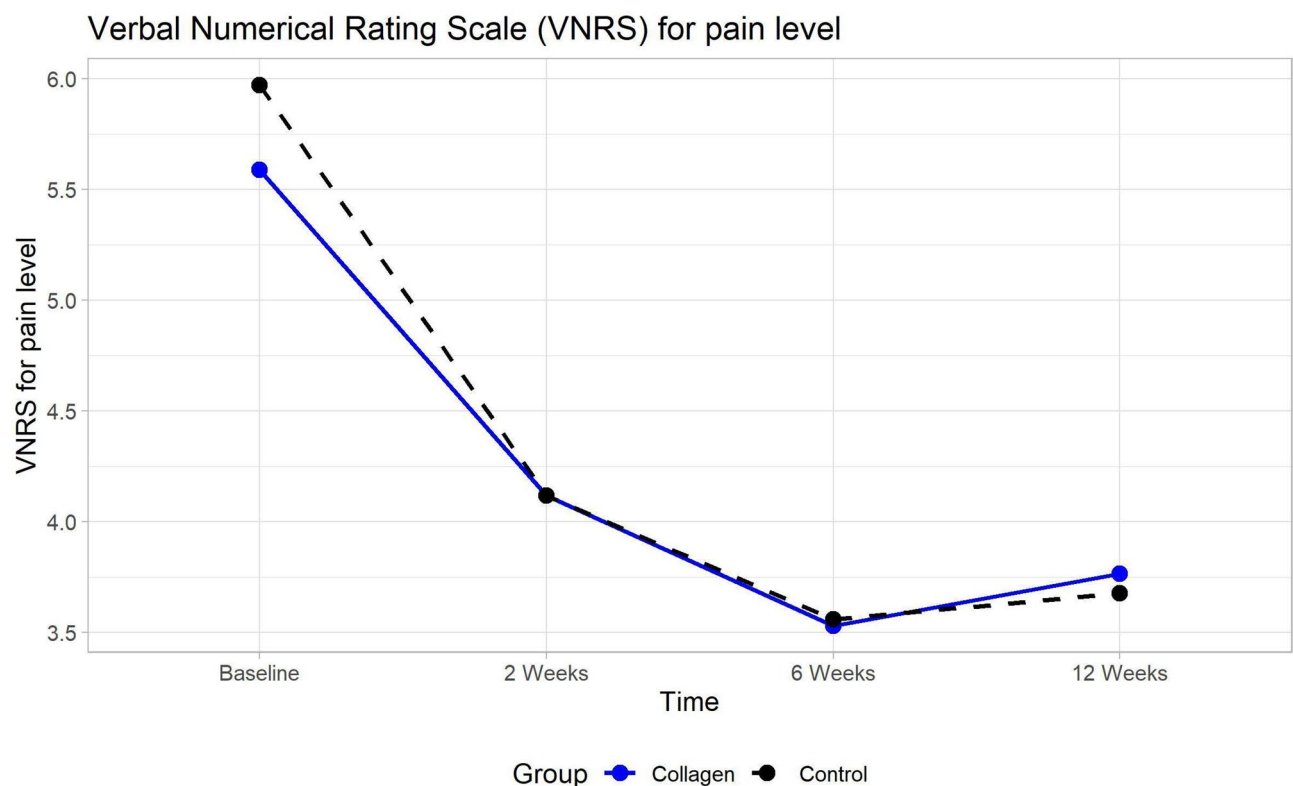


Fig. 2. Verbal numerical rating scale (VNRS) for pain level.

The KOOS assessment showed similar baseline scores between groups across all subscales (Pain, Symptoms, Activities of Daily Living, Sports/Recreation, and Quality of Life; $p > 0.05$). Both groups demonstrated improvements in KOOS scores over time; however, no significant between-group differences were observed throughout the study period. (Table 3)

Both groups showed a decreasing trend in rescue medication consumption over time. At 2 weeks, the collagen group had a lower median consumption compared with that in the control group (1 [IQR, 6] vs. 4 [IQR, 10] tablets), although this difference was not significant ($p = 0.135$). The median consumption continued to decrease at 6 weeks (collagen: 0 [IQR, 5]; control: 0.5 [IQR, 10]; $p = 0.48$) and reached similar levels at 12 weeks (collagen: 0 [IQR, 4]; control: 0 [IQR, 5]; $p = 0.518$). (Table 2)

The patient satisfaction scores were consistently high in both groups throughout the study period. The collagen group reported median satisfaction scores of 8 (IQR, 3), 9 (IQR, 2), and 8.5 (IQR, 1.3) at 2, 6, and 12 weeks, respectively. Similarly, the control group reported median scores of 7 (IQR, 3), 8 (IQR, 3), and 9 (IQR, 2), respectively. No significant differences were observed between the groups at any time point ($p = 0.227, 0.355$, and 0.126 at 2, 6, and 12 weeks, respectively). No adverse effects were reported from taking either collagen or placebo.

		Baseline	2 weeks	6 weeks	12 weeks
Pain	Collagen	69 (20)	77 (17)	80 (26)	83 (15)
	Control	63 (23)	75 (13)	78 (9)	83 (18)
	p value	0.325	0.887	0.912	0.863
	95% CI	– 3, 11	– 6, 6	– 8, 6	– 5, 6
Symptoms	Collagen	71 (23)	82 (15)	82 (25)	86 (19)
	Control	75 (13)	81 (18)	82 (16)	84 (15)
	p value	0.887	0.820	0.725	0.839
	95% CI	– 7, 7	– 4, 7	– 7, 4	– 4, 7
Activities of daily living	Collagen	65 (30)	67 (24)	69 (23)	72 (18)
	Control	60 (18)	65 (16)	69 (17)	75 (21)
	p value	0.102	0.443	0.768	0.610
	95% CI	– 1, 16	– 4, 9	– 8, 6	– 9, 6
Sports and recreation	Collagen	15 (25)	20 (26)	23 (26)	23 (28)
	Control	15 (20)	15 (16)	18 (15)	25 (21)
	p value	0.216	0.446	0.354	1
	95% CI	0, 10	– 5, 10	– 5, 10	– 10, 10
Quality of life	Collagen	35 (25)	50 (12)	50 (15)	69 (25)
	Control	38 (25)	50 (14)	50 (25)	69 (25)
	p value	0.841	0.923	0.421	0.804
	95% CI	– 6, 6	– 6, 6	– 12, 0	– 6, 0

Table 3. Comparison of Knee injury and Osteoarthritis Outcome Score (KOOS) between collagen and control groups. *IQR* interquartile range, *CI* confidence interval.

Discussion

The management of knee OA primarily focuses on symptom control and functional improvement. Recently, nutritional supplements, particularly collagen-based products, have garnered interest because of their potential role in cartilage metabolism and promising results in previous studies¹⁶. Our study evaluated the efficacy of combined UC-II and hydrolysed collagen supplementation in patients with knee OA. Although both intervention and placebo groups demonstrated significant improvements in pain and functional outcomes over the 12-week study period, no significant differences were observed between the groups across all measured parameters, including pain intensity, functional scores, rescue medication consumption, and patient satisfaction.

These findings differ from those of several previous studies that have examined UC-II supplementation alone. Sadigursky et al.¹⁷ reported significant improvements in both Western Ontario and McMaster University Arthritis Index (WOMAC) Pain and visual analogue scale (VAS) scores at 30 and 90 d in the UC-II group compared with those in the placebo group. Similarly, Lugo et al.¹² observed superior outcomes in the UC-II group across all WOMAC subscales, including significant reductions in pain and VAS scores on day 180.

However, our results aligned more closely with those of Costa et al.,¹⁸ who also found no significant differences between the UC-II and placebo groups.

Regarding functional outcomes, our findings contradicted those of several previous studies. We observed no significant improvements in KOOS scores between the groups; however, Sadigursky et al.¹⁷ demonstrated significant enhancements in WOMAC function scores and SF-12 physical components by day 90. Similarly, Lugo et al.¹² reported superior outcomes across all functional parameters in the UC-II group on day 180.

When comparing our results with those of previous studies on hydrolysed collagen, we found similar disparities. While earlier research by Trč and Bohmová¹⁹, Benito-Ruiz et al.,¹⁴ and Bruyère et al.²⁰ reported superior pain relief with collagen hydrolysate compared with control interventions, our combined supplementation approach did not yield similar benefits. Notably, Trč and Bohmová¹⁹ demonstrated that compared with glucosamine sulphate, enzymatic hydrolysed collagen resulted in significant VAS improvements, while Benito-Ruiz et al.¹⁴ observed marked improvements in both VAS and WOMAC scores, particularly among severe OA cases.

The discrepancy between our findings and those of previous studies may be attributed to several factors including supplement formulation variations, dosing regimens, study durations, and patient populations.

Notably, previous trials of individual collagen products such as Lugo et al.¹² Sadigursky et al.¹⁷ and Benito-Ruiz et al.¹⁴ reported significant improvements in pain and function with intervention durations of 90 days to 6 months. In contrast, our study assessed a combined formulation over a shorter 12-week period. It is possible that the clinical benefits of collagen supplementation, particularly those related to cartilage metabolism and repair, may require longer durations to become detectable. The relatively short timeframe in our study may therefore have limited the ability to observe meaningful differences between groups.

The absence of superior outcomes in the combined supplementation group may suggest a lack of synergistic interaction between UC-II and hydrolysed collagen. These two compounds may even compete for absorption or bioactivity pathways. Additionally, the dosages used in this study (20 mg UC-II and 480 mg hydrolysed collagen) may be suboptimal when administered together. The fixed dosing may not reflect the ideal ratio or threshold needed for synergistic effects. Future studies should explore optimal dosing strategies, timing of administration,

and potential pharmacokinetic interactions between these agents to determine whether combined use can offer additive or synergistic clinical benefits.

Our study has several limitations. First, the 12-week follow-up period may have been insufficient to detect the long-term effects of collagen supplementation on cartilage metabolism and clinical outcomes, as the biological effects of nutritional interventions often require extended durations to manifest. Second, the relatively small sample size may have resulted in insufficient statistical power to detect small but potentially clinically meaningful effects. A post hoc power analysis using the observed between-group difference in pain reduction (0.08 on the VNRS) and an estimated standard deviation of 2.0 yielded a very small effect size (Cohen's $d = 0.04$), corresponding to only 5.3% power with our current sample size ($n = 34$ per group). Based on conventional estimates for nutraceutical trials, a small-to-moderate effect size (Cohen's $d = 0.3$) would require approximately 235 participants per group to achieve 90% power at a 0.05 significance level. Therefore, the null findings should be interpreted cautiously, as the study may have been underpowered to detect subtle benefits. Future research with larger sample sizes is warranted to more definitively assess efficacy.

Third, the study did not include objective biomarkers, imaging data, or pharmacokinetic assessments, which limits our ability to evaluate potential disease-modifying effects or confirm treatment adherence. Future studies should consider incorporating serum or synovial biomarkers, pharmacokinetic analyses, or imaging modalities such as MRI to better understand the biological impact of collagen supplementation and to verify compliance. Additionally, the single-centre recruitment and predominance of moderate knee OA cases (Kellgren–Lawrence grade 2–3) may limit the generalisability of our findings to patients with varying disease severities or from different clinical settings. Although rescue medication consumption was monitored, individual variations in pain threshold and medication usage behaviours could have influenced analgesic patterns. Nonetheless, the randomised double-blind design and comprehensive assessment of both objective and patient-reported outcomes provide robust support for our primary conclusions.

Conclusion

This randomised double-blind trial demonstrated that combined supplementation with UC-II and hydrolysed collagen over 12 weeks provided no additional benefit compared with that of placebo in patients with knee OA. Although both groups showed improvements, the lack of between-group differences suggested that these changes may reflect the natural disease course or placebo effect. Longer-term multicentre studies are needed to fully evaluate the potential benefits of combined collagen supplementation in specific OA populations.

Data availability

The datasets generated during this current study are available from the corresponding author upon reasonable request.

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Author contributions

VY: Study design, data collection, statistical analysis, and writing the paper; PT, TH : Data collection; CA : Study design, statistical analysis, and writing the paper; KI: Corresponding author, study design, data collection, and writing the paper. All authors have read and approved the final manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Ethics approval

This study was approved by the Ethics Committee and Institutional Review Board of the Faculty of Medicine, Prince of Songkla University.

Consent to participate

Informed consent was obtained from all individual participants included in the study.

Additional information

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